End-to-End Loan Default Analysis & Portfolio Risk Assessment

Professional business case report for GitHub portfolio: Python data preparation, feature engineering, machine learning, SQL business analysis, and Power BI dashboarding.

2,000	60.9%	INR 94.0 Cr	AUC 0.8845
clean loan records	baseline default rate	portfolio exposure	ML model performance

Executive takeaway

The project converts raw borrower data into a decision-ready credit risk workflow. It identifies risk patterns, creates predictive default probabilities, answers business questions in SQL, and communicates portfolio risk through a three-page Power BI dashboard.

Tools used

Stage	Tools / Output
Data preparation	Jupyter Notebook, Python, Pandas, NumPy
Analysis	EDA charts, risk score, feature engineering
Prediction	Logistic Regression, AUC/ROC, default probability
Business layer	SQL sections with KPIs, segmentation, loss estimates
Visualization	Power BI executive dashboard with 3 pages and 44 visuals

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1. Business Problem and Project Objective

Business challenge

Loan defaults can create major losses for banks and lending companies. A risk team needs more than a static default count; it needs a workflow that finds risky borrower profiles, estimates financial exposure, and highlights where policy or monitoring should change.

Project objective

• Understand the structure and quality of raw loan application data.

• Clean and transform the data into an analysis-ready table.

• Engineer risk indicators such as loan-to-income ratio, EMI burden, credit tier, income mismatch, and composite risk score.

• Train a basic machine learning model to generate default probabilities.

• Use SQL to translate model outputs into business KPIs and decision recommendations.

• Build Power BI dashboard pages for executives, risk analysts, and operations teams.

Business questions answered

Question	Why it matters
Which borrowers are most likely to default?	Supports early risk flagging and manual review.
Which attributes drive portfolio risk?	Guides underwriting and credit policy.
How large is the estimated loss exposure?	Helps executives prioritize risk controls.
Which dashboards are needed by stakeholders?	Makes insights usable by non-technical users.

2. End-to-End Analytics Pipeline

Workflow overview

The project follows a practical analytics lifecycle, starting from raw CSV ingestion and ending with business-ready dashboard views. Each layer is connected to the next so that cleaning, prediction, SQL analysis, and visualization use a consistent dataset.

Step	Layer	Deliverable
1	Raw data collection	Raw CSV with borrower demographics, loan details, credit attributes, and default label.
2	Python notebook	Data audit, cleaning, transformation, feature engineering, EDA, and ML modeling.
3	Clean prediction CSV	Final dataset with engineered columns and ML default probabilities.
4	SQL analysis	Business KPIs, segmentation, financial impact, trend analysis, and automated recommendations.
5	Power BI dashboard	Three report pages for executive monitoring, borrower risk analysis, and financial impact.

GitHub-ready value

This structure is strong for recruiters because it proves complete ownership of a data project: not only dashboarding, but also data preparation, modeling, SQL, and business storytelling.

3. Data Audit, Cleaning, and Preparation

Raw dataset snapshot

2,010	29	6	9 fields
raw rows	raw columns	duplicates removed	with missing values

Data quality issues detected

Field	Missing count	Missing %
employment status	140	7.0%
dti ratio	120	6.0%
credit score	107	5.3%
months employed	100	5.0%
education	81	4.0%
annual income inr	81	4.0%
loan purpose	61	3.0%
num late payments	61	3.0%
interest rate	41	2.0%

Cleaning actions performed

- Removed duplicate records and dropped the duplicated loan ID field.
- Converted credit score from text to numeric and validated the acceptable range.
- Corrected impossible values such as invalid ages, negative loan amounts, and extreme DTI ratios.
- Fixed income entry errors where values appeared to be recorded in the wrong unit.
- Standardized gender and loan purpose fields.
- Parsed application dates into year, month, and quarter fields.
- Imputed numeric nulls with median values for robust modeling.

4. Feature Engineering and Risk Framework

Why feature engineering matters

The raw dataset did not directly contain all business risk signals needed for underwriting decisions. New features were created to capture affordability, credit quality, stability, and model-ready borrower risk.

Feature / Indicator	Business meaning
credit_tier	Groups CIBIL scores into understandable risk bands.
monthly_income	Converts annual income into monthly affordability context.
estimated_emi	Approximates monthly repayment burden.
emi_to_income_ratio	Measures EMI pressure relative to income.
loan_to_income_ratio	Compares loan exposure against annual income.
income_mismatch_flag	Flags mismatch between stated and analyzed income.
stable_employment	Indicates employment stability from months employed.
risk_score_raw	Composite risk score from credit, DTI, late payments, bankruptcies, and loan burden.
ml_default_probability	Model-driven probability of default.

Rule-based risk segmentation

Risk category	Loan count	Default rate	Avg ML probability	Portfolio Cr
High Risk	113	100.0%	98.2%	7.35
Low Risk	657	32.0%	30.0%	24.94
Medium Risk	1229	72.7%	74.0%	61.65
Very High Risk	1	100.0%	99.8%	0.06

5. Exploratory Data Analysis Highlights

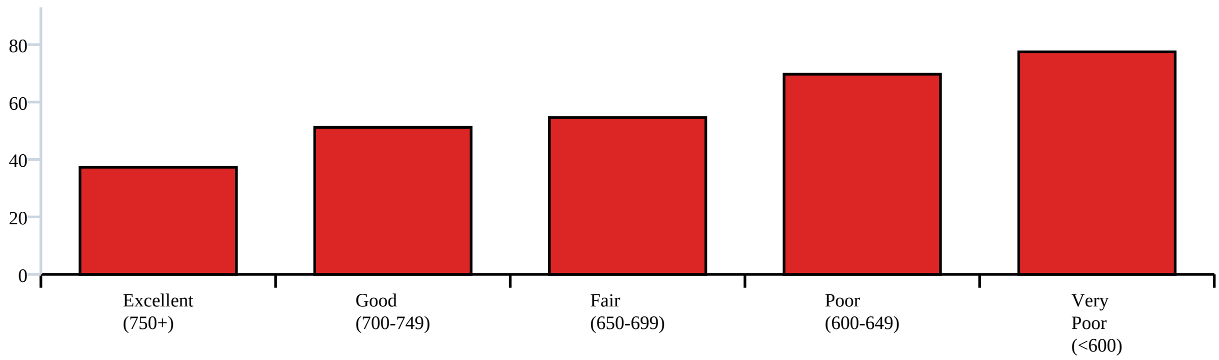
Portfolio snapshot

1,218	782	INR 4.7 L	INR 5.11 L
defaulted loans	performing loans	average loan	average income

Credit tier and loan grade patterns

Default risk rises materially as credit quality deteriorates. The Very Poor credit tier shows a 77.5% default rate compared with 37.3% in the Excellent tier. Loan grade also follows a clear risk ladder from A/B to G.

Default rate by CIBIL tier (%)



Loan grade	Loan count	Default rate	Avg interest rate
A	87	35.6%	11.19%
B	132	37.9%	11.62%
C	230	47.4%	12.02%
D	367	53.1%	12.36%
E	501	63.3%	12.62%
F	491	72.5%	13.49%
G	192	83.3%	13.73%

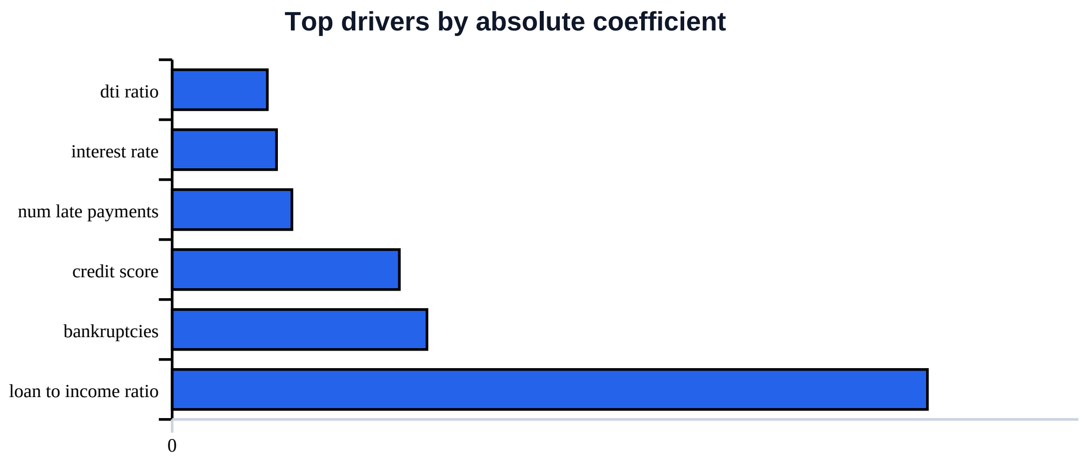
6. Predictive Modeling and Model Interpretation

Modeling approach

A logistic regression model was trained as a transparent baseline classifier for predicting loan default. The model uses borrower credit quality, affordability, repayment behavior, income, loan amount, employment stability, and engineered risk features.

0.8845	78.5%	81.6%	83.6%
ROC-AUC	accuracy	default precision	default recall

Top model drivers



Business interpretation

The model confirms that loan-to-income ratio, bankruptcies, credit score, late payments, interest rate, and DTI ratio are important risk signals. This aligns with practical credit risk logic: repayment burden, past credit behavior, and affordability pressure are central to default risk.

7. Prediction Output and Decision Support

ML risk flag performance

The model probability was converted into an ML risk flag using a 0.60 threshold. This creates a practical review list that a credit team can use for manual validation or stricter approval criteria.

1,075	85.7%	921	75.6%
loans flagged by ML	default rate in flagged group	defaults captured	capture rate

ML probability band	Loan count	Observed default rate
<30%	414	16.2%
30-50%	346	43.4%
50-70%	334	55.7%
70%+	906	90.0%

Decision design

Segment	Suggested action
Low probability	Standard approval process with routine monitoring.
Medium probability	Additional income and affordability checks.
High probability	Manual underwriting review and stricter loan terms.
Very high probability	Reject or require strong mitigating evidence.

8. SQL Business Analysis and Power BI Dashboard

SQL analytics layer

The SQL file contains 10 labeled analysis sections and 31 SELECT statements. It turns the clean prediction dataset into business KPIs, default segmentation, financial impact estimates, geography trends, demographic views, fraud indicators, and automated loan decision recommendations.

SQL section	Purpose
Database overview	Total loans, defaults, default rate, average loan, average credit score, portfolio value.
Default rate analysis	Risk by credit tier, loan grade, employment, loan purpose, and borrower attributes.
Financial impact	Portfolio exposure, estimated default loss, and model-driven savings logic.
Early warning indicators	Income mismatch, late payments, bankruptcies, high DTI, and other risk signals.
Executive KPI summary	BI-ready outputs for dashboard cards and decision views.

Power BI dashboard structure

Page	Visuals	Business user
Executive Summary	14	Executive / risk team
Borrower Risk Analysis	15	Risk and operations team
Financial Impact	15	Risk and operations team

Dashboard insight

The Power BI report contains three pages and 44 visuals: Executive Summary, Borrower Risk Analysis, and Financial Impact. The dashboard makes model outputs and SQL KPIs easier to consume for non-technical stakeholders.

9. Recommendations, Limitations, and GitHub Usage

Strategic recommendations

• Tighten underwriting rules for borrowers with high loan-to-income ratios, repeated late payments, or bankruptcies.

• Use the ML probability score as a review trigger rather than a standalone approval decision.

• Monitor high-exposure states and loan grades where default rates are materially elevated.

• Create a weekly risk review using the Power BI dashboard for defaults, exposure, and financial impact.

• Combine rule-based risk scoring with ML outputs to capture more risky loans than a strict rule-based approach alone.

Limitations and future improvements

• The model is a transparent baseline model; future work can compare Random Forest, XGBoost, and calibrated probability models.

• The dashboard is portfolio-focused; future work can add borrower-level drill-through pages and alerts.

• The data contains personal identifiers; public GitHub versions should remove or mask applicant names and PAN numbers.

• A production pipeline should validate data quality automatically before model scoring and BI refresh.

Recommended GitHub folders

Folder	Purpose
notebooks/	Python data cleaning, EDA, feature engineering, and ML model notebook.
data/	Raw sample and clean prediction CSV, with sensitive fields removed if public.
sql/	Business analysis SQL queries.
powerbi/	Power BI PBIX file.
screenshots/	Dashboard images so recruiters can preview the work quickly.
reports/	This full business case report PDF.
presentation/	Recruiter-ready PowerPoint and PDF presentation.

Final conclusion

This project is a complete data analytics case study. It demonstrates data cleaning, feature engineering, EDA, model building, SQL business analysis, and Power BI storytelling - the core skills required for junior data analyst, BI analyst, SQL analyst, and credit risk analytics roles.